

> PRACTICE EXERCISE

1. Frequency polygon can be drawn after drawing
 - (a) ogive
 - (b) bar chart
 - (c) histogram
 - (d) None of these
2. An ogive is used to determine
 - (a) mean
 - (b) median
 - (c) GM
 - (d) HM
3. The mid-value of a class interval is 42. If the class size is 10, then the upper and lower limits of the class are
 - (a) 37.5 and 47.5
 - (b) 47 and 37
 - (c) 37 and 47
 - (d) 47.5 and 37.5
4. The actual lower class limits of the following classes 10-19, 20-29, 30-39 and 40-49 are
 - (a) 9.5, 19, 29 and 39.5
 - (b) 10, 20, 30 and 40
 - (c) 9.5, 19.5, 29.5 and 39.5
 - (d) 18.5, 28.5, 38.5 and 48.5
5. If the mean of five observations $x, x + 2, x + 4, x + 6$ and $x + 8$ is 11, then the mean of first three observations is
 - (a) 9
 - (b) 11
 - (c) 13
 - (d) None of these
6. The combined mean of three groups is 12 and the combined mean of first two groups is 3. If the first, second and third groups have 2, 3 and 5 items respectively, then mean of third group is
 - (a) 10
 - (b) 21
 - (c) 12
 - (d) 18
7. 10 is the mean of a set of 7 observations and 5 is the mean of a set of 3 observations. The mean of the combined set is given by
 - (a) 15
 - (b) 10
 - (c) 8.5
 - (d) 7.5
8. In a class of 50 students, 10 have failed and their average marks are 28. The total marks obtained by the entire class are 2800. The average marks of those who have passed are
 - (a) 43
 - (b) 53
 - (c) 63
 - (d) 70
9. Which of the following statements about the median is true?
 - (a) It is not affected by extreme values
 - (b) It can be found even, if some items are not known
 - (c) It is useful when the data cannot be measured quantitatively
 - (d) All of the above
10. The middle item of the series arranged in ascending or descending order is called
 - (a) mean
 - (b) median
 - (c) mode
 - (d) standard deviation
11. The following pie chart shows the marks obtained by a student in an examination who scored 540 marks in all. The subject in which the student scored 108 marks is

Subject	Angle (°)
Mathematics	90°
English	63°
Social Science	72°
Science	75°
Hindi	60°

 - (a) Science
 - (b) Hindi
 - (c) English
 - (d) Social Science
12. A distribution consists of three components with frequencies 45, 40 and 15 having their means 2, 2.5 and 2 respectively. The mean of the combined distribution is
 - (a) 2.1
 - (b) 2.2
 - (c) 2.3
 - (d) 2.4
13. If the values $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots, \frac{1}{n}$ occur at frequencies 1, 2, 3, 4, 5, 6, ..., n respectively, in a frequency distribution, then the mean is
 - (a) 1
 - (b) n
 - (c) $\frac{1}{n}$
 - (d) $\frac{2}{n+1}$
14. If the geometric mean of three observations 40, 50 and x is 10, then the value of x is
 - (a) $\frac{1}{2}$
 - (b) 4
 - (c) 6
 - (d) 2
15. Suppose, X is some statistical variable with mean μ . Let $x_1, x_2, \dots, x_n$ be its deviations from mean with the respective frequencies $f_1, f_2, \dots, f_n$. What is the value of the sum $x_1f_1 + x_2f_2 + \dots + x_nf_n$?
 - (a) 0
 - (b) 1
 - (c) μ
 - (d) $\mu + 1$
16. Which one of the following statements is not correct with reference to a histogram?
 - (a) Frequency curve is obtained by joining the mid-points of the top of the adjacent rectangles with smooth curves
 - (b) Histogram is drawn for continuous data
 - (c) The height of the bar is proportional to the frequency of that class
 - (d) Mode of the distribution can be obtained from the histogram

17.

X	0	1	2	3	4
Frequency	4	f	9	g	4

The table above gives the frequency distribution of a discrete variable X with two missing frequencies. If the total frequency is 25 and the arithmetic mean of X is 2, then what is the value of the missing frequency f ?

- (a) 4 (b) 5 (c) 6 (d) 7

- 18.** The yield of paddy per plot of one acre were obtained from a number of plots from two different districts in a state and are summarized in the following table.

Yield of paddy per plot in quintals	Number of plots in district A	Number of plots in district B
38.0-41.0	25	14
41.0-44.0	36	29
44.0-47.0	59	35
47.0-50.0	30	54
50.0-53.0	25	41

Which one of the following statements is correct?

- (a) The mode for district A is higher than the mode for district B
 (b) The mode for district B is higher than the mode for district A
 (c) Both the distributions are symmetric
 (d) Both the distributions have the same mean
- 19.** Average score of 50 students in a class is 44. Later on it was found that the score 23 was incorrectly recorded as 73. The correct average score is
- (a) 42 (b) 43 (c) 45 (d) 46
- 20.** In the "less than" type of ogive the cumulative frequency is plotted against
- (a) the lower limit of the concerned class interval
 (b) the upper limit of the concerned class interval
 (c) the mid value of the concerned class interval
 (d) any value of the concerned class interval
- 21.** A, B and C are three sets of values of x
 A : 2, 3, 7, 1, 3, 2, 3
 B : 7, 5, 9, 12, 5, 3, 8
 C : 4, 4, 11, 7, 2, 3, 4
- Select the correct statement from among the following
- (a) Mean of A is equal to mode of C
 (b) Mean of C is equal to median of B
 (c) Median of B is equal to mode of A
 (d) Median, mean and mode of A are same
- 22.** A student obtains 75%, 80% and 85% marks in three subjects. If the marks of any other subject are added, then their average cannot be less than
- (a) 60% (b) 65% (c) 70% (d) 80%

- 23.** If the mean and median of a set of numbers are 8.9 and 9 respectively, then the mode will be
- (a) 7.2 (b) 8.2 (c) 9.2 (d) 10.2

- 24.** If every number of a finite set is increased by any number k , then the measure of central tendency should also increase by k . Which one of the following measures of central tendency does not have this property?

- (a) Arithmetic mean
 (b) Median
 (c) Mid-range i.e. the arithmetic mean of the largest and the smallest numbers
 (d) Geometric mean

- 25.** If the median of the distribution (arranged in ascending order) 1, 3, 5, 7, 9, x , 15, 17, 19, 21 is 10, what is the value of x ?

- (a) 11 (b) 13 (c) $9 < x < 15$ (d) $9 \leq x \leq 15$

- 26.** If the width of each of the ten classes in a frequency distribution is 2.5 and the lower class boundary of the lowest class is 5.1, then the upper class boundary of the highest class is

- (a) 30.1 (b) 30 (c) 31.1 (d) 27.56

- 27.** The average value of the median of 2, 8, 3, 7, 4, 6, 7 and the mode of 2, 9, 3, 4, 9, 6, 9 is

- (a) 9 (b) 8 (c) 7.5 (d) 6

- 28.** When 10 is subtracted from each of the given observations, the mean is reduced to 60%. If 5 is added to all the given observations, the mean will be

- (a) 25 (b) 30 (c) 60 (d) 65

- 29.** A data has highest value 120 and the lowest value 71. A frequency distribution in descending order with seven classes is to be constructed.

The limits of the second class interval shall be

- (a) 77 and 78 (b) 78 and 85
 (c) 85 and 113 (d) 113 and 120

- 30.** If mean of y and $\frac{1}{y}$ is M , then what is the mean of y^3 and $\frac{1}{y^3}$?

- (a) $\frac{M(M^2 - 3)}{3}$ (b) M^3
 (c) $M^3 - 3$ (d) $M(4M^2 - 3)$

- 31.** For the following frequency distribution

Class interval	0-5	5-10	10-15	15-20	20-25	25-30
Frequency	10	15	30	80	40	20

If m is the value of mode, then which one of the following is correct?

- (a) $5 < m < 10$ (b) $10 < m < 15$
 (c) $15 < m < 20$ (d) $20 < m < 25$

- 32.** Square diagrams are drawn to represent the following data

Country	Pakistan	India	Myanmar	China
Labour Production (in ₹)	36	81	25	100

Using the scale $1 \text{ cm}^2 = ₹ 25$ what is the length of the representative square for India?

- (a) 1.8 cm (b) 1.2 cm (c) 1 cm (d) 2 cm
- 33.** The standard deviation of 7, 9, 11, 13, 15 is
(a) 2.4 (b) 2.5 (c) 2.7 (d) 2.82
- 34.** The total number of cellphones sold for Motorola, Samsung and Sony was 45664. The number of cellphones sold for these companies were in the ratio 3:5:8 respectively. If these data were shown on a pie chart, calculate the angle represented by the number of cellphones sold by Motorola.
(a) 75° (b) 67.5° (c) 70° (d) 74.5°
- 35.** If the population figures are given for each State of India, then the data can be classified as
(a) qualitative (b) quantitative
(c) chronological (d) geographical
- 36.** Which one of the following represents statistical data?
(a) The names of all owners of shops located in a shopping complex
(b) A list giving the names of all States of India
(c) A list of all European countries and their respective capital cities
(d) The volume of a rainfall in certain geographical area, recorded every month for 24 consecutive months
- 37.** Prime numbers are the numbers which comes in the table of 1 and itself only.
I. The mean of first seven prime numbers is greater than their median.
II. Mean is always greater than median.
Select the correct option from the options given below
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 38.** Consider the following statements in respect of a histogram
I. The histogram consists of vertical rectangular bars with a common base such that there is no gap between consecutive bars.
II. The height of the rectangle is determined by the frequency of the class it represents.
Which of the statements given above is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 39.** Consider the following statements in respect of histogram
I. Histogram is an equivalent graphical representation of the frequency distribution.
II. Histogram is suitable for continuous random variables, where the total frequency of an interval is evenly distributed over the interval.
Which of the statements given above is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 40.** Examples of data are given below
I. Information on households collected by an investigator by door-to-door visits.
II. Data on the percentage of literates, sex-wise, for the different districts of a state collected from records of the census of India.
III. General information about families, collected by telephonic interviews.

Which one of the following in respect of the above is correct?

- (a) I and II are primary data (b) I and III are primary data
(c) II and III are primary data (d) I, II and III are primary data

- 41.** Consider the following types of data

- I. Marks of students who appeared for a test of 100 marks.
II. Collar sizes of 200 shirts sold in a week.
III. Monthly incomes of 250 employees of a factory.

For which of the above data, mode is a suitable measure of central tendency?

- (a) I and II (b) Only II (c) I and III (d) All of these

- 42.** The cumulative frequency curve of a frequency distribution with 6 classes and total frequency 60 is a straight line. Consider the following statements

- I. The first and the last classes have a frequency of 10 each.
II. Both the middle classes have a total frequency of 30.
III. The frequency distribution does not have a mode.

Which of the statements given above are correct?

- (a) I and II (b) I and III (c) II and III (d) All of these

- 43.** State which of the following variables are discrete?

- I. Number of children in a family.
II. Wages of workers.
III. The ages of students.

- IV. Weights of a set of a students.

Select the correct answer using the codes given below

- (a) I and II (b) I, II and III
(c) All of these (d) None of these

Directions (Q. Nos. 44-45) The number of goals scored by a soccer team in a season were as follows:

Number of goals	Number of matches
0	3
1	6
2	2
3	x
4	5

44. If the mode is 1, the largest possible value of x is
(a) 2 (b) 5 (c) 3 (d) 6
45. If the median is 2, the smallest possible value of x is
(a) 2 (b) 5 (c) 3 (d) 6

Directions (Q. Nos. 46-47) The itemwise expenditure of a non-government organisation for the year 2008-2009 is given below.

Item	Expenditure (in ₹ lakh)
Salary of employees	6
Social welfare activities	7
Office contingency	3
Vehicle maintenance	4
Rent and hire charges	2.5
Miscellaneous expenses	1.5

The above data are represented by a pie diagram.

46. What is the sectorial angle of the largest sector?
(a) 120° (b) 105° (c) 90° (d) 85°
47. What is the difference in the sectorial angles of the largest and the smallest sectors?
(a) 90° (b) 85° (c) 82.5° (d) 77.5°

Directions (Q. Nos. 48-49) The following table gives the frequency distribution of life length in hours of 100 electric bulbs having median life 20 h.

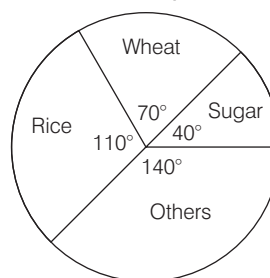
Life of bulbs (in h)	Number of bulbs
8-13	7
13-18	x
18-23	40
23-28	y
28-33	10
33-38	2

48. What is the missing frequency of x ?
(a) 31 (b) 27 (c) 24 (d) 14
49. What is the missing frequency of y ?
(a) 27 (b) 24 (c) 14 (d) 11

Directions (Q. Nos. 50-52) The arithmetic mean, geometric mean and median of 6 positive numbers a, a, b, b, c, c where $a < b < c$ are $\frac{7}{3}, 2, 2$ respectively.

50. What is the sum of the squares of all the six numbers?
(a) 40 (b) 42 (c) 45 (d) 48
51. What is the value of c ?
(a) 1 (b) 2 (c) 3 (d) 4
52. What is the mode?
(a) 1 (b) 2
(c) 1, 2 and 4 (d) None of these

Directions (Q. Nos. 53-55) Study the following pie diagram and answer the questions.



53. The percentage of wheat production is
(a) 35% (b) 70% (c) 19.44% (d) 20%
54. If the sugar production is 9000 kg, then the wheat production is
(a) 20000 kg (b) 24000 kg
(c) 15750 kg (d) None of these
55. If the total production is 180000 kg, the difference in sugar and wheat production is
(a) 10000 (b) 15000 (c) 20000 (d) None of these

Directions (Q. Nos. 56-58) Consider the following frequency distribution

Class interval	1-4	4-7	7-10	10-13	13-16	16-19
Frequency	6	30	40	16	4	4

56. What is the median of the data?
(a) 9.4 (b) 7.68 (c) 8.05 (d) 8.32
57. What is modal size of the given data?
(a) 8.14 (b) 7.88 (c) 7.62 (d) 8.48
58. The value of (Mean + Mode - 2 Median) is equal to
(a) 0 (b) 0.2 (c) 0.4 (d) 0.1

PREVIOUS YEARS' QUESTIONS

59. Let the observations at hand be arranged in increasing order. Which one of the following measures will not be affected when the smallest and the largest observations are removed?

- (a) Mean (b) Median
(c) Mode (d) Standard deviation

2012 I

- 60.** Consider the following statements :
- I. The data collected by the investigator to be used by himself are called primary data.
- II. The data obtained from government agencies are called secondary data.

Which of the statement(s) given above is/are correct? **☑ 2012 I**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 61.** Which one among the following statements is correct? **☑ 2012 I**

- (a) Simple bar diagrams are those diagrams which show two characteristics of the data
(b) In pie diagrams, all the items are converted into angles
(c) A bar diagram is one in which data are shown in terms of bars
(d) Bar diagrams present data through length and breadth

- 62.** Consider the following distribution :

Value of the variable	1	2	3	4	5
Frequency	3	f	6	5	3

For what value of f , is the arithmetic mean of the above distribution 3.1? **☑ 2012 I**

- (a) 2 (b) 3 (c) 4 (d) 5

- 63.** Which of the following pair(s) is/are correctly matched?

- I. Weight of a person : Continuous variable
II. Educational qualification of the person : Attribute
(a) Only I (b) Only II **☑ 2012 II**
(c) Both I and II (d) Neither I nor II

- 64.** The mean of 100 values is 45. If 15 is added to each of the first forty values and 5 is subtracted from each of the remaining sixty values, then the new mean becomes **☑ 2012 II**

- (a) 45 (b) 48 (c) 51 (d) 55

- 65.** Which one of the following relations for the numbers 10, 7, 8, 5, 6, 8, 5, 8 and 6 is correct?

- (a) Mean = Median (b) Mean = Mode **☑ 2012 II**
(c) Mean > Median (d) Mean > Mode

- 66.** In histogram the width of the bars is proportional to **☑ 2012 II**

- (a) Frequency (b) Number of classes
(c) Class interval (d) None of these

- 67.** Which one of the following statements is correct? **☑ 2013 I**

- (a) A frequency polygon is obtained by connecting the corner points of the rectangles in a histogram
(b) A frequency polygon is obtained by connecting the mid-points of the tops of the rectangles in a histogram
(c) A frequency polygon is obtained by connecting the corner points of the class intervals in a histogram
(d) None of the above

- 68.** Consider the following statements :

- I. A frequency distribution condenses the data and reveals its important features.
II. A frequency distribution is an equivalent representation of original data.

Which of the statement(s) given above is/are correct? **☑ 2013 I**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Directions (Q. Nos. 69-71) Read the following information carefully to answer the questions that follow. **☑ 2013 I**

In a frequency distribution having class intervals 0-10, 10-20, 20-30 and 30-40 the respective frequencies are x , $x+8$, $x-2$ and $x-4$ and the arithmetic mean of the distribution is 17.8.

- 69.** The value of x is
(a) 3 (b) 6 (c) 8 (d) 12

- 70.** The median lies in which one of the following class intervals?

- (a) 0-10 (b) 10-20 (c) 20-30 (d) 30-40

- 71.** The modal class is

- (a) 0-10 (b) 10-20 (c) 20-30 (d) 30-40

- 72.** The mean of 7 observations is 7. If each observation is increased by 2, then the new mean is **☑ 2013 I**

- (a) 12 (b) 10 (c) 9 (d) 8

- 73.** There are 45 male and 15 female employees in an office. If the mean salary of the 60 employees is ₹ 4800 and the mean salary of the male employees is ₹ 5000, then the mean salary of the female employees is **☑ 2013 I**

- (a) ₹ 4200 (b) ₹ 4500 (c) ₹ 5600 (d) ₹ 6000

Directions (Q. Nos. 74-75) Read the following information carefully and answer the questions given below. **☑ 2013 II**

The median of the following distribution is 14.4 and the total frequency is 20.

Class interval	0-6	6-12	12-18	18-24	24-30
Frequency	4	x	5	y	1

- 74.** What is x equal to?

- (a) 4 (b) 5 (c) 6 (d) 7

- 75.** What is the relation between x and y ?

- (a) $2x = 3y$ (b) $3x = 2y$ (c) $x = y$ (d) $2x = y$

- 76.** If m is the mean of p, q, r, s, t, u and v , then what is $(p-m) + (q-m) + (r-m) + (s-m) + (t-m) + (u-m) + (v-m)$ equal to? **☑ 2013 II**

- (a) 0 (b) s
(c) $\frac{(p+v)}{2}$ (d) None of these